Bonigi Srisanth

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.27
Narayana Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh (BIEAP))	2024	96.70%
Mythilee E M H School	Class X (Board of Secondary Education, Andhra Pradesh (BSEAP))	2022	95.83%

SKILLS

Programming Languages: C, C++, Python, SQL

Databases: SQLite

Tools & Software: Git, GitHub, MATLAB, Jupyter Notebook, LaTeX

Machine Learning: scikit-learn, pandas, NumPy, sentence-transformers, Seaborn, XGBoost, TensorFlow, OpenCV, Matplotlib

Soft Skills: Leadership, Communication, Adaptability, Teamwork, Event Execution, Crisis Management, Dedication

PROJECTS

GPT-Style Decoder-Only Transformer

Personal Project | Python, PyTorch, Transformers, Multi-Head Attention

- Implemented a decoder-only GPT Transformer architecture from scratch in PyTorch, building token and positional embeddings, multi-head masked self-attention, feed-forward (MLP) blocks, residual connections, layer normalization, and output projection without using pre-built Transformer modules.
- Engineered the complete autoregressive training pipeline, including causal attention masking, scaled dot-product attention, vocabulary tokenization, batched sequence generation, cross-entropy loss optimization, backpropagation, and mini-batch training for next-token prediction.
- Developed a modular and extensible Transformer framework, implementing configurable attention heads, embedding dimensions, context windows, and stacked decoder blocks to enable efficient end-to-end language modeling and text generation.

Real-Time Face Recognition Attendance System | Computer Vision & Machine Learning

Personal Project | Python, OpenCV, SQL, Face-Recognition

- Developed a real-time facial recognition system using OpenCV and dlib-based face-recognition, generating 128-dimensional embeddings for automated identity verification from live video.
- Architected a SQLite database with duplicate-detection logic for accurate, one-entry-per-day attendance logging and scalable user enrollment.
- Optimized the detection-recognition pipeline via frame downscaling and configurable match-tolerance thresholds, and built a CSV export module for seamless reporting integration.

System Call-Aware Multi-Level Feedback Queue Scheduler

Course Project - Operating Systems | C++, xv6, Kernel Programming

- Extended the xv6 kernel by implementing custom system calls for process management, system call accounting, and parent-child process tracking.
- Implemented a System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ) scheduler featuring dynamic priority adjustment, round-robin scheduling, and starvation prevention through global priority boosting.
- Developed kernel-level scheduling statistics and comprehensive user-space test suites to benchmark CPU-bound and interactive workloads, improving scheduler fairness and responsiveness.

MiniDB - In-Memory Key-Value Database

Personal Project | C++, Data Structures, Database Systems

- Built a Redis-inspired in-memory key-value database in C++ with SET, GET, DELETE, EXISTS, and prefix search using a dynamically resizable hash table for O(1) average lookups.
- Implemented O(1) LRU eviction and TTL expiration using a Doubly Linked List + Hash Map and a Min-Heap, enabling efficient cache management and automatic key expiration.
- Engineered a Trie-based prefix index with dynamic rehashing, separate chaining for collision resolution, and a modular object-oriented architecture for scalable performance.

RELEVANT COURSES

Computer Science courses: Introduction to Programming, Data Structures and its Applications, Operating Systems I & II.

Mathematics: Introduction to Probability, Statistics, Complex Variables, Transform Techniques, Differential Equations,Elementary Linear Algebra, Calculus I & II

Computational Courses: Fundamentals of Scientific Computing, Computer aided Numerical Methods 1 & 2, Computational Methods in Material Science, Electronic Structure and Atomic Modeling of Materials

POSITIONS OF RESPONSIBILITY

Sponsorship Coordinator | Elan & nVision, IIT Hyderabad's Annual Techno Cultural Fest

- Managed and mentored a team of 10+ Deputy Coordinators, driving sponsorship outreach, stakeholder engagement, and partnership acquisition.
- Secured 10L+ in sponsorships through negotiations with corporate partners, achieving 20%+ of the targeted sponsorship revenue during the early stages of the fundraising campaign.

Events Team Coordinator | Inter IIT Sports Meet

- Coordinated large-scale event operations for 3,000+ participants from 23 IITs, managing logistics, venue readiness, scheduling, and real-time issue resolution across multiple sporting events.
- Collaborated with officials, volunteers, and organizing committees to streamline inter-team communication, optimize resource allocation, and ensure uninterrupted event execution.

Events Head - Diesta 2026, IIT Hyderabad

- Inter Departmental Sports and Cultural fest of IIT Hyderabad.

Operations Coordinator - Finance & Consulting Club, IIT Hyderabad

- Coordinated club operations and cross-team initiatives, streamlining event execution and supporting finance and consulting-focused activities.

EXTRACURRICULAR

Active Chess Player under the National Sports Organization (NSO), IIT Hyderabad